

Branching and boundary tests

Project: Python readiness portfolio

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Boolean conditions

Without opening the old lesson, explain one key idea from Boolean conditions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Values variables and types

Without opening the old lesson, explain one key idea from Values variables and types. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the tested path for Branching and boundary tests. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Branching and boundary tests. Proposed explanation: “Every condition in the chain always runs”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Trace branch order

Predict the action for 30, then trace 29 and -1. Explain why validity is checked before the lamp decision.

```
reading = 30

if not 0 <= reading <= 100:
    action = "invalid"

elif reading < 30:
    action = "on"

else:
```

```
    action = "off"

print(action)
```

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the chain

Describe the purpose of if, elif and else in the worked duration sorter. Explain why print(label) is outside the indented branches.

My reasoning and evidence

6. W2 Trace the tested path

For -1, 12 and 13, record checked conditions, skipped conditions and the final label. Run each as a fresh script after predicting.

My reasoning and evidence

7. W3 Repair the overwritten result

Trace the two-independent-if example for -1. Identify which assignment overwrites the earlier label and make the smallest repair.

My reasoning and evidence

8. W4 Build a new sorter independently

Use an integer tiles count. Label a negative value Invalid, zero Empty, 1 through 6 Small and values above 6 Large. Write an ordered if elif else script and predict -1, 0, 1, 6 and 7. Include one print after the chain.

My reasoning and evidence

9. W5 Choose revealing regression tests

A repair changes the small-count condition from < 6 to ≤ 6 . Name the case that exposes the original error and two different kinds of earlier cases to rerun.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled branching and boundary tests evidence sheet to Python readiness portfolio. Show how you can trace the selected branch in an if elif else chain. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Branching and boundary tests

Check invalid values before branching. A toy shelf label uses fewer than five books as "small", otherwise "large", but negative counts are invalid.

Python 3: `if count<0: label="invalid"; elif count<5: label="small"; else: label="large"`. Read as indented branches.

Branching and boundary tests

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Check validity

For count=-2, the first condition selects invalid.



2. Check small boundary

For count=4, validity passes and count<5 selects small.



3. Handle equality

For count=5, both earlier conditions fail, so large is selected.

CHANGE ONE THING

Remove the first negative-count branch.

Explain the changed result using the matching lesson.

Trace -2, 4 and 5 through the branches. Explain why the negative check comes first and what label -2 gets if it is removed.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.